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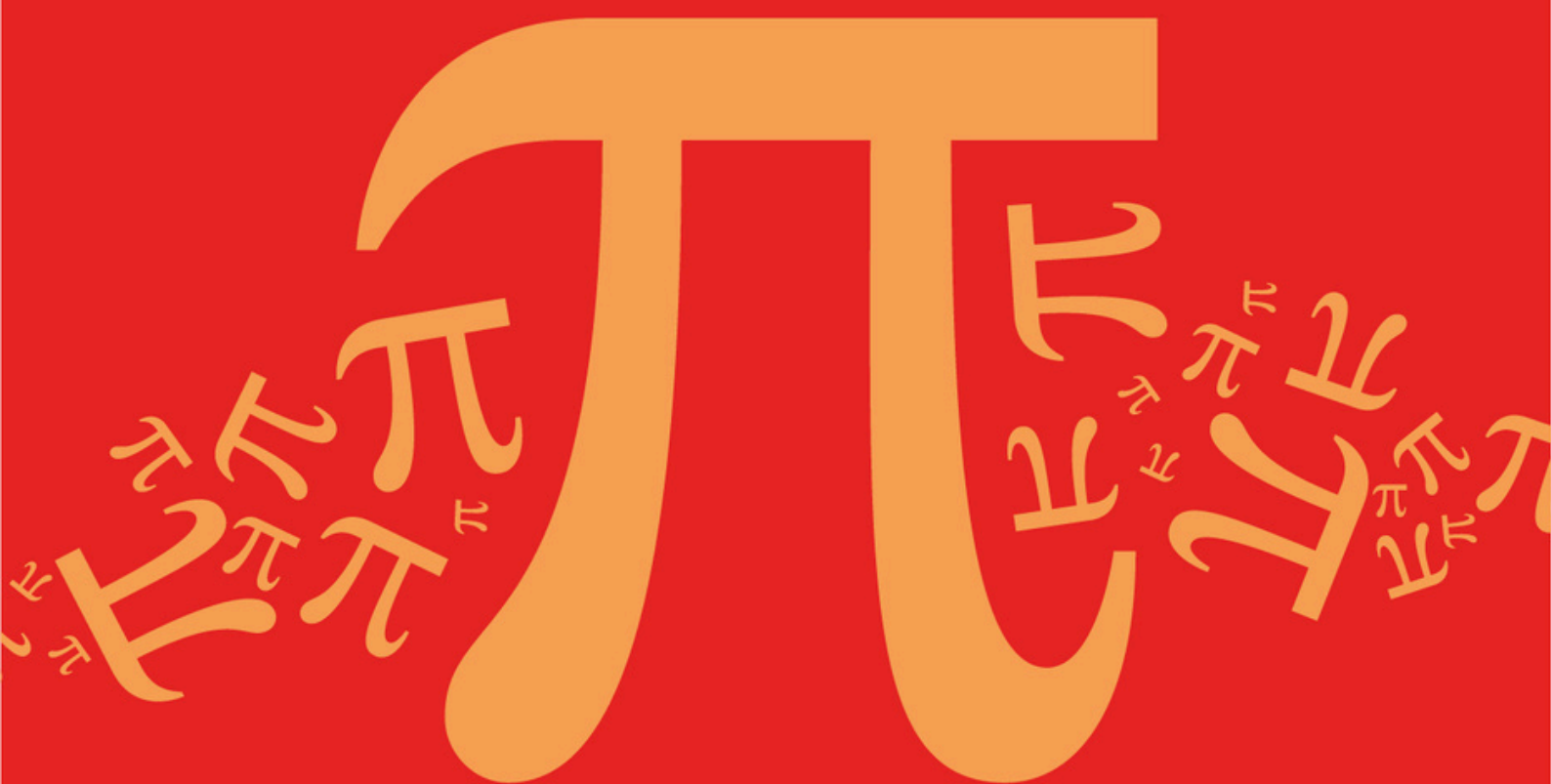
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AQA A Level Further Mathematics (7367)



Topic

C: Matrices

Sub topic

C10: Eigenvalues and Eigenvectors

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

C: Matrices

Sub topic

C10: Eigenvalues and Eigenvectors

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

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Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

$$\mathbf{M} = \begin{bmatrix} -1 & 2 & -1 \\ 2 & 2 & -2 \\ -1 & -2 & -1 \end{bmatrix}$$

- (a) Given that 4 is an eigenvalue of $\mathbf{M}$, find a corresponding eigenvector.

(3)

- (b) Given that $\mathbf{MU} = \mathbf{UD}$, where $\mathbf{D}$ is a diagonal matrix, find possible matrices for $\mathbf{D}$ and $\mathbf{U}$.

(8)

(Total 11 marks)

Q2.

The matrix $\mathbf{M}$ is defined by

$$\mathbf{M} = \begin{bmatrix} -a & 0 & a \\ 0 & 6 & 0 \\ a & 0 & 2 \end{bmatrix}$$

where a is a real number. The distinct eigenvalues of $\mathbf{M}$ are λ_1 , λ_2 and λ_3 with corresponding eigenvectors $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$.

- (a) Given that $\mathbf{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, find λ_1 .

(2)

- (b) Given that $\mathbf{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$, find the value of a .

(3)

- (c) Given that $\lambda_3 = -6$, find a possible eigenvector $\mathbf{v}_3$.

(3)

- (d) The matrix $\mathbf{M}$ can be expressed as $\mathbf{UDU}^{-1}$, where $\mathbf{D}$ is a diagonal matrix.

Write down possible matrices $\mathbf{D}$ and $\mathbf{U}$.

(3)

(Total 11 marks)

Q3.

The matrix $\mathbf{M}$ is given by

$$\mathbf{M} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{bmatrix}$$

- (a) Show that $\lambda = 2$ is an eigenvalue for $\mathbf{M}$, and find the other two eigenvalues. (5)
- (b) Find an eigenvector that corresponds to $\lambda = 2$. (3)
- (c) The matrix $\mathbf{N}$ is given by

$$\mathbf{N} = \begin{bmatrix} 2 & 2 & -3 \\ 2 & 2 & 3 \\ -3 & 3 & 3 \end{bmatrix}$$

- (i) Show that $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is an eigenvector for $\mathbf{N}$, and find the corresponding eigenvalue. (2)
- (ii) Hence state one eigenvector for the matrix $\mathbf{MN}$, and find the corresponding eigenvalue. (3)

(Total 13 marks)

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Q4.

- (a) Find the eigenvalues and corresponding eigenvectors of the matrix $\mathbf{M} = \begin{bmatrix} 4 & 5 \\ 5 & 4 \end{bmatrix}$. (6)
- (b) The plane transformation T is given by the matrix $\mathbf{M}$. Write down the coordinates of the invariant point of T . (1)
- (Total 7 marks)

Q5.

The matrix $\mathbf{A} = \begin{bmatrix} k & 1 & 2 \\ 2 & k & 1 \\ 1 & 2 & k \end{bmatrix}$, where k is a real constant.

- (a) (i) Show that there is a value of k for which

$$\mathbf{A} \mathbf{A}^T = m \mathbf{I}$$

where m is a rational number to be determined and $\mathbf{I}$ is the 3×3 identity matrix.

- (6)
- (ii) Deduce the inverse matrix, $\mathbf{A}^{-1}$, of $\mathbf{A}$ for this value of k . (1)
- (b) (i) Find $\det \mathbf{A}$ in terms of k . (2)
- (ii) In the case when $\mathbf{A}$ is singular, find the integer value of k and show that there are no other possible real values of k . (3)
- (iii) Find the value of k for which $\lambda = 7$ is a real eigenvalue of $\mathbf{A}$. (2)
- (Total 14 marks)

Q6.

Let $\mathbf{Y} = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 3 & 1 \\ 1 & 1 & 3 \end{bmatrix}$.

- (a) Show that 4 is a repeated eigenvalue of $\mathbf{Y}$, and find the other eigenvalue of $\mathbf{Y}$. (7)
- (b) For each eigenvalue of $\mathbf{Y}$, find a full set of eigenvectors. (5)
- (c) The matrix $\mathbf{Y}$ represents the transformation T .

Describe the geometrical significance of the eigenvectors of $\mathbf{Y}$ in relation to T .

(3)
(Total 15 marks)

Q7.

- (a) (i) Find the eigenvalues and corresponding eigenvectors of $\mathbf{A} = \begin{bmatrix} 1 & 3 \\ -2 & 8 \end{bmatrix}$. (6)
- (ii) Hence write down each of the matrices $\mathbf{U}$, $\mathbf{D}$ and $\mathbf{U}^{-1}$ such that $\mathbf{A} = \mathbf{UDU}^{-1}$, where $\mathbf{D}$ is a diagonal matrix. (4)
- (b) A 2×2 matrix $\mathbf{M}$ has distinct real eigenvalues γ and μ , with corresponding eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$.
- (i) By considering the diagonalised form of $\mathbf{M}$, determine the eigenvalues of $\mathbf{M}^3$. (2)

- (ii) Write down the eigenvectors of $\mathbf{M}^3$

(1)

(Total 13 marks)

Q8.

(a) It is given that $\Delta = \begin{vmatrix} 16-q & 5 & 7 \\ -12 & -1-q & -7 \\ 6 & 6 & 10-q \end{vmatrix}$.

- (i) By using row operations on the first two rows of Δ , show that $(4 - q)$ is a factor of Δ .

(2)

- (ii) Express Δ as the product of three linear factors.

(4)

(b) It is given that $\mathbf{M} = \begin{bmatrix} 16 & 5 & 7 \\ -12 & -1 & -7 \\ 6 & 6 & 10 \end{bmatrix}$.

- (i) Verify that $\begin{bmatrix} 2 \\ 5 \\ -7 \end{bmatrix}$ is an eigenvector of $\mathbf{M}$ and state its corresponding eigenvalue.

(3)

- (ii) For each of the other two eigenvalues of $\mathbf{M}$, find a corresponding eigenvector.

(7)

- (c) The transformation T has matrix $\mathbf{M}$. Write down cartesian equations for any one of the invariant lines of T .

(2)

(Total 18 marks)

Q9.

The transformation T is represented by the matrix $\mathbf{M}$ with diagonalised form

$$\mathbf{M} = \mathbf{U} \mathbf{D} \mathbf{U}^{-1}$$

where $\mathbf{U} = \begin{bmatrix} 4 & -1 \\ 1 & 3 \end{bmatrix}$ and $\mathbf{D} = \begin{bmatrix} 27 & 0 \\ 0 & 1 \end{bmatrix}$.

- (a) (i) State the eigenvalues, and corresponding eigenvectors, of $\mathbf{M}$.

(4)

- (ii) Find a cartesian equation for the line of invariant points of T .

(2)

- (b) Write down $\mathbf{U}^{-1}$, and hence find the matrix $\mathbf{M}$ in the form

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

where a, b, c and d are integers.

(5)

- (c) By finding the element in the first row, first column position of $\mathbf{M}^n$, prove that

$$4 - 3^{3n+1} + 1$$

is a multiple of 13 for all positive integers n .

(5)

(Total 16 marks)

Q10.

- (a) Given that -1 is an eigenvalue of the matrix $\mathbf{M} = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 3 \end{bmatrix}$, find a corresponding eigenvector.

(3)

- (b) Determine the other two eigenvalues of $\mathbf{M}$, expressing each answer in its simplest surd form.

(8)

(Total 11 marks)

Q11.

The plane transformation T is defined by

$$T: \begin{bmatrix} x' \\ y' \end{bmatrix} = \mathbf{M} \begin{bmatrix} x \\ y \end{bmatrix}$$

where $\mathbf{M} = \begin{bmatrix} -1 & 4 \\ -1 & 3 \end{bmatrix}$.

- (a) Evaluate $\det \mathbf{M}$ and state the significance of this answer in relation to T .

(2)

- (b) Find the single eigenvalue of $\mathbf{M}$ and a corresponding eigenvector. Describe the geometrical significance of these answers in relation to T .

(5)

- (c) Show that the image of the line $y = \frac{1}{2}x + k$ under T is $y' = \frac{1}{2}x' + k$.

(3)

(Total 10 marks)

Q12.

The 2×2 matrix $\mathbf{M}$ has an eigenvalue 3, with corresponding eigenvector $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, and a second eigenvalue -3 , with corresponding eigenvector $\begin{bmatrix} 1 \\ 4 \end{bmatrix}$.

The diagonalised form of $\mathbf{M}$ is $\mathbf{M} = \mathbf{U} \mathbf{D} \mathbf{U}^{-1}$.

- (a) (i) Write down suitable matrices $\mathbf{D}$ and $\mathbf{U}$, and find $\mathbf{U}^{-1}$. (4)
- (ii) Hence determine the matrix $\mathbf{M}$. (3)
- (b) Given that n is a positive integer, use the result $\mathbf{M}^n = \mathbf{U} \mathbf{D}^n \mathbf{U}^{-1}$ to show that:
- (i) when n is even, $\mathbf{M}^n = 3^n \mathbf{I}$;
- (ii) when n is odd, $\mathbf{M}^n = 3^{n-1} \mathbf{M}$. (6)
- (Total 13 marks)

Q13.

The matrix $\mathbf{T}$ has eigenvalues 2 and -2 , with corresponding eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 3 \end{bmatrix}$ respectively.

- (a) Given that $\mathbf{T} = \mathbf{U} \mathbf{D} \mathbf{U}^{-1}$, where $\mathbf{D}$ is a diagonal matrix, write down suitable matrices $\mathbf{U}$, $\mathbf{D}$ and $\mathbf{U}^{-1}$. (3)
- (b) Hence prove that, for all **even** positive integers n ,

$$\mathbf{T}^n = f(n) \mathbf{I}$$

where $f(n)$ is a function of n , and $\mathbf{I}$ is the 2×2 identity matrix.

(5)
(Total 8 marks)

Q14.

The non-singular matrix $\mathbf{M} = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}$.

- (a) (i) Show that

$$\mathbf{M}^2 + 2\mathbf{I} = k\mathbf{M}$$

for some integer k to be determined.

(3)

- (ii) By multiplying the equation in part (a)(i) by $\mathbf{M}^{-1}$, show that

$$\mathbf{M}^{-1} = a\mathbf{M} + b\mathbf{I}$$

for constants a and b to be found.

(3)

- (b) (i) Determine the characteristic equation of $\mathbf{M}$ and show that $\mathbf{M}$ has a repeated eigenvalue, 1, and another eigenvalue, 2.

(6)

- (ii) Give a full set of eigenvectors for each of these eigenvalues.

(5)

- (iii) State the geometrical significance of each set of eigenvectors in relation to the transformation with matrix $\mathbf{M}$.

(3)

(Total 20 marks)

Q15.

Find the eigenvalues and corresponding eigenvectors of the matrix $\begin{bmatrix} 7 & 12 \\ 12 & 0 \end{bmatrix}$.

(Total 6 marks)

Q16.

A transformation T of three-dimensional space is given by the matrix $\mathbf{W} = \begin{bmatrix} 3 & -1 & 1 \\ 2 & 0 & 2 \\ -1 & 1 & 1 \end{bmatrix}$.

- (a) (i) Evaluate $\det \mathbf{W}$, and describe the geometrical significance of the answer in relation to T .

(2)

- (ii) Determine the eigenvalues of $\mathbf{W}$.

(6)

- (b) The plane H has equation $\mathbf{r} \cdot \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 0$.

- (i) Write down a cartesian equation for H .

(1)

- (ii) The point P has coordinates (a, b, c) . Show that, whatever the values of a , b and c , the image of P under T lies in H .

(4)

(Total 13 marks)

Q17.

- (a) Find the eigenvalues and corresponding eigenvectors of the matrix

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 5 & 4 \end{bmatrix}$$

(6)

- (b) (i) Write down a diagonal matrix $\mathbf{D}$, and a suitable matrix $\mathbf{U}$, such that

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{U}^{-1}$$

(2)

- (ii) Write down also the matrix $\mathbf{U}^{-1}$.

(1)

- (iii) Use your results from parts (b)(i) and (b)(ii) to determine the matrix $\mathbf{X}^5$ in

the form $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, where a, b, c and d are integers.

(3)

(Total 12 marks)

Q18.

The matrix $\mathbf{M} = \begin{bmatrix} 1 & -1 & 1 \\ 3 & -3 & 1 \\ 3 & -5 & 3 \end{bmatrix}$,

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- (a) Given that $\mathbf{u} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ are eigenvectors of $\mathbf{M}$, find the eigenvalues corresponding to $\mathbf{u}$ and $\mathbf{v}$.

(5)

- (b) Given also that the third eigenvalue of $\mathbf{M}$ is 1, find a corresponding eigenvector.

(6)

- (c) (i) Express the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ in terms of $\mathbf{u}$ and $\mathbf{v}$.

(1)

- (ii) Deduce that $\mathbf{M}^n \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \lambda^n \mathbf{u} + \mu^n \mathbf{v}$, where λ and μ are scalar constants whose values should be stated.

(4)

- (iii) Hence prove that, for all positive **odd** integers n ,

$$\mathbf{M}^n \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2^n \\ 0 \\ 2^n \end{bmatrix}$$

(3)

(Total 19 marks)

Q19.

For real numbers a and b , with $b \neq 0$ and $b \neq \pm a$, the matrix

$$\mathbf{M} = \begin{bmatrix} a & b+a \\ b-a & -a \end{bmatrix}$$

- (a) (i) Show that the eigenvalues of $\mathbf{M}$ are b and $-b$.

(3)

- (ii) Show that $\begin{bmatrix} b+a \\ b-a \end{bmatrix}$ is an eigenvector of $\mathbf{M}$ with eigenvalue b .

(2)

- (iii) Find an eigenvector of $\mathbf{M}$ corresponding to the eigenvalue $-b$.

(2)

- (b) By writing $\mathbf{M}$ in the form $\mathbf{UDU}^{-1}$, for some suitably chosen diagonal matrix $\mathbf{D}$ and corresponding matrix $\mathbf{U}$, show that

$$\mathbf{M}^{11} = b^{10} \mathbf{M}$$

(7)

(Total 14 marks)

Q20.

The matrix $\mathbf{M}$ is given by $\begin{bmatrix} 2 & 7 \\ 4 & k \end{bmatrix}$, where k is a constant. It is given that the vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector of $\mathbf{M}$.

- (a) Show that $k = 5$ and find the eigenvalue corresponding to the eigenvector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

(3)

- (b) Determine the second eigenvalue of $\mathbf{M}$ and find a corresponding eigenvector.

(5)

- (c) Write down matrices $\mathbf{U}$ and $\mathbf{D}$, having integer elements, such that $\mathbf{M}$ can be expressed in the diagonalised form $\mathbf{M} = \mathbf{UDU}^{-1}$.

(2)

- (d) Write down the matrix $\mathbf{U}^{-1}$.

(1)

- (e) The matrix $\mathbf{M}^{2n}$, for positive integers n , is such that

$$\mathbf{M}^{2n} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

for integers a, b, c and d . Show that

$$a = p \times 4^n + q \times 81^n$$

where p and q are rational numbers to be determined.

(4)

(Total 15 marks)



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